

Team Goodnight — Check-in 2
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Update:

Our core research question remains unchanged: we are examining how individual stress and district-level environmental features relate to sleep quality across major Chinese cities. Since Check 1, however, several substantive updates have strengthened our data structure, environmental measures, and workflow.

1. Environmental pipeline finalized and standardized.

We expanded our remote sensing pipeline (NDVI from MODIS and nighttime lights from VIIRS) from a single-city test case to all five cities. Each district (ADM3) now includes:

- Mean NDVI (June 30 – August 31, 2023)
- Mean VIIRS nighttime radiance (June 30 – August 31, 2023)

The time window was refined to a clearly defined summer period to avoid mixing long-term environmental averages with cross-sectional survey outcomes. Extraction and aggregation are fully reproducible through a single Python script.

2. Integration of sports facility density.

A major expansion since Check 1 was the addition of district-level sports facility density (facilities per 100,000 residents), compiled and cleaned by Rebecca. This built-environment measure introduces a theoretically meaningful neighborhood variable that may moderate stress-related sleep outcomes. Incorporating this required administrative name harmonization and district-level merge validation across all five cities.

3. Division of responsibilities clarified.

Shoshana: remote sensing extraction, district-level aggregation, repository structure, documentation.
Rebecca: survey cleaning, facility density integration, exploratory visualization, and preparation for modeling.

4. Challenges addressed.

We resolved district name inconsistencies across datasets, validated merges through spot checks, and began addressing skew in the sleep outcome distribution.

Data Analysis and Visualization Plan

We have begun exploratory data analysis on the merged dataset containing:

- pss4_total (stress)
- sleep_quality
- NDVI
- nighttime lights
- sports facility density

Preliminary Findings

1. Sleep quality distribution.

The histogram shows a right-skewed distribution, with most observations between 1–5 and a long upper tail. This suggests meaningful variation but potential non-normality.

2. Stress and sleep.

Mean sleep quality by stress level (with 95% confidence intervals) shows a gradual upward trend. However, the slope is moderate rather than steep, and differences between adjacent stress levels are small. Confidence intervals widen at higher stress levels. This indicates a moderate gradient rather than a dramatic shift.

3. Facility density quartiles.

Boxplots comparing sleep quality across facility density quartiles show similar medians across groups, with slightly greater variability in higher-density quartiles. There is no sharp discontinuity between low and high density. Differences appear mild to moderate rather than strong main effects.

Required: Specific Data Analysis Decision

Based on EDA, we will estimate an exploratory linear regression model:

$$\text{sleep_quality} \sim \text{pss4_total} + \text{facility_per_100k_pop} + \text{pss4_total} \times \text{facility_per_100k_pop}$$

Outcome: sleep_quality

Individual-level predictor: pss4_total

District-level predictor: facility_per_100k_pop

Interaction: stress $\times$ facility density

Rationale: Stress shows a moderate association with sleep, while facility density alone shows mild differences. The interaction term allows us to test whether neighborhood infrastructure moderates the stress–sleep relationship. This moves beyond simple main effects and aligns directly with our research question.

Required: Structural Improvement for Visualization

To move from exploratory to polished presentation:

- Present NDVI, nightlights, and facility density on comparable scales.
- Standardize y-axis scales across panels
- Facet key plots instead of pooling all cities
- Annotate sample sizes in higher stress bins to clarify widening confidence intervals.

These changes will allow clearer cross-city comparisons and reduce overinterpretation of pooled patterns.

Focus Area

Our focus area is examining how individual stress interacts with district-level environmental conditions (green space, nighttime light exposure, and sports facility density) to shape variation in sleep quality across urban districts.

Next Steps

- Conduct residual diagnostics due to skewed outcome distribution.
- Evaluate multicollinearity among NDVI, nightlights, and facility density.
- Run regression models with interaction terms
- Finalize city-faceted visualizations for presentation.